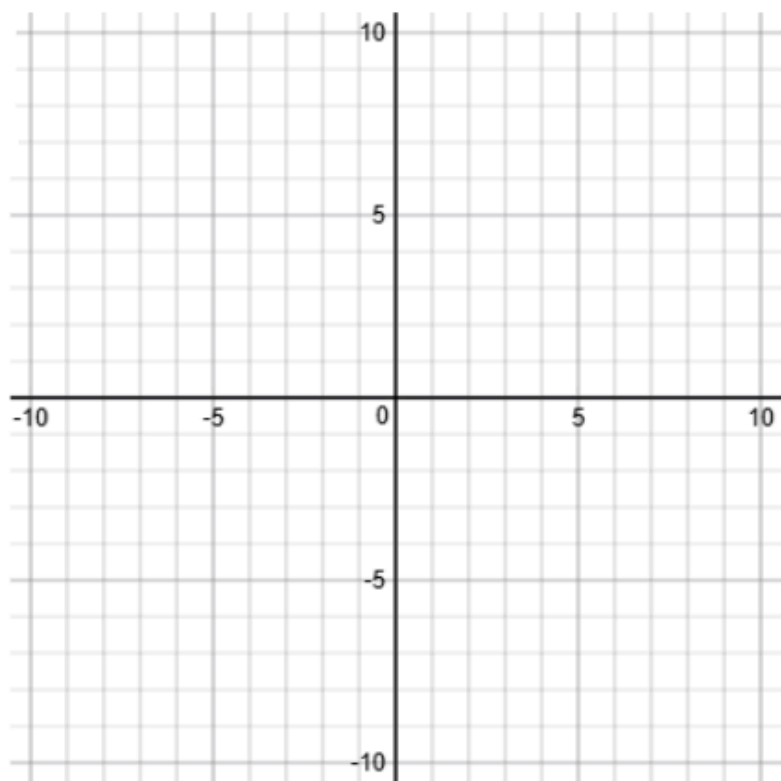


Calculus I Lesson 03

$$1) f(x) = \begin{cases} 3 - x, & x \neq 2 \\ 1, & x = 2 \end{cases}$$

$$a) \text{ Graph } f(x) = \begin{cases} 3 - x, & x \neq 2 \\ 1, & x = 2 \end{cases}$$



b) Complete the following tables.

x	$f(x)$
1.5	
1.9	
1.999	
1.999999	

x	$f(x)$
2.5	
2.1	
2.0001	
2.0000001	

b) Complete the following tables.

x	$f(x)$
1.5	
1.9	
1.999	
1.999999	

x	$f(x)$
2.5	
2.1	
2.0001	
2.0000001	

c) $\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 2 from the left

$\lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

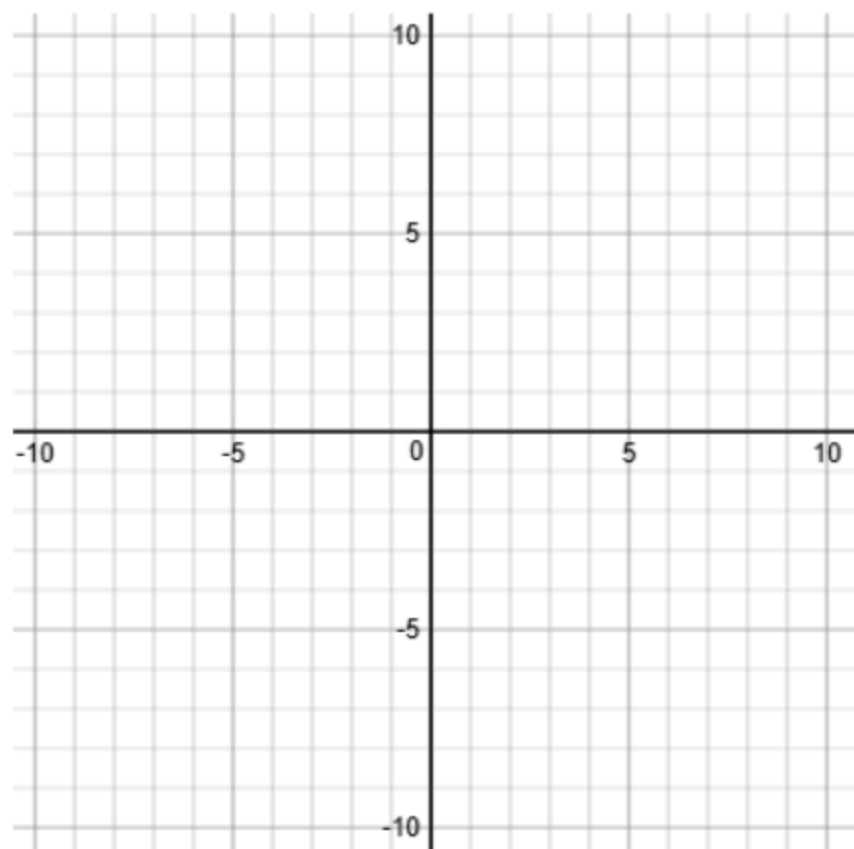
limit of $f(x)$ as x approaches 2 from the right

$\lim_{x \rightarrow 2} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 2

$$2) f(x) = \begin{cases} 2 - x^2, & x \leq 1 \\ 3x, & x > 1 \end{cases}$$

$$\text{a) Graph } f(x) = \begin{cases} 2 - x^2, & x \leq 1 \\ 3x, & x > 1 \end{cases}$$



b) Complete the following tables.

x	$f(x)$
0.5	
0.9	
0.999	
0.999999	

x	$f(x)$
1.5	
1.1	
1.0001	
1.0000001	

c) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches 1 from the left

$\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{1cm}} ?$

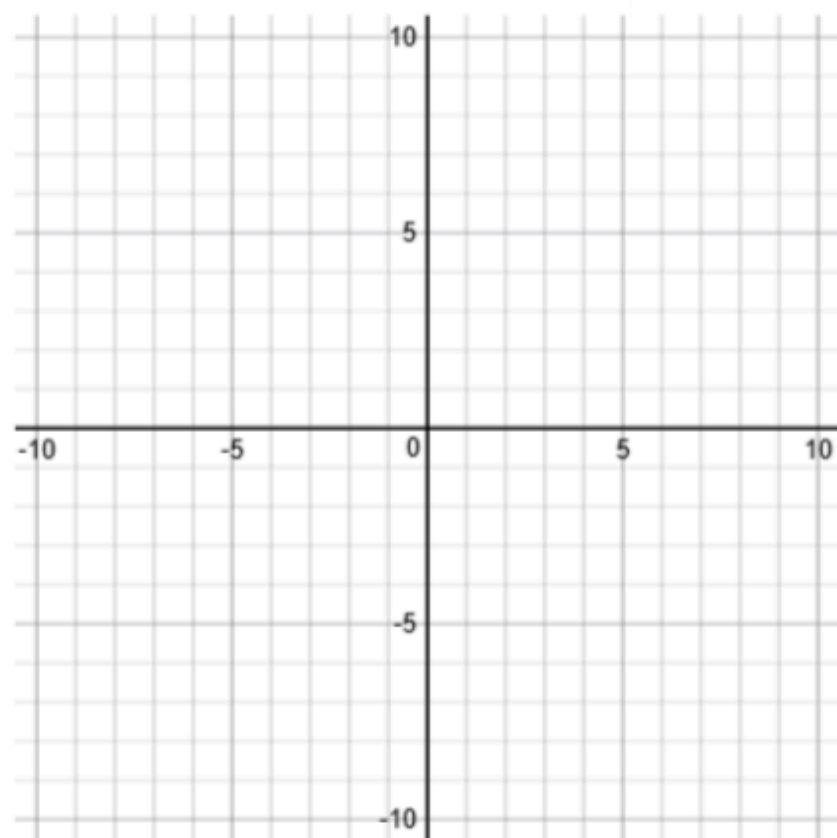
limit of $f(x)$ as x approaches 1 from the right

$\lim_{x \rightarrow 1} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches 1

$$3) f(x) = \begin{cases} x^2, & x \leq 1 \\ 4 - 3x, & 1 < x < 3 \\ 2, & x \geq 3 \end{cases}$$

$$a) f(x) = \begin{cases} x^2, & x \leq 1 \\ 4 - 3x, & 1 < x < 3 \\ 2, & x \geq 3 \end{cases}$$



$$b) \lim_{x \rightarrow 1^-} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 1 from the left

$$\lim_{x \rightarrow 1^+} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 1 from the right

$$\lim_{x \rightarrow 1} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 1

c) $\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 3 from the left

$\lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

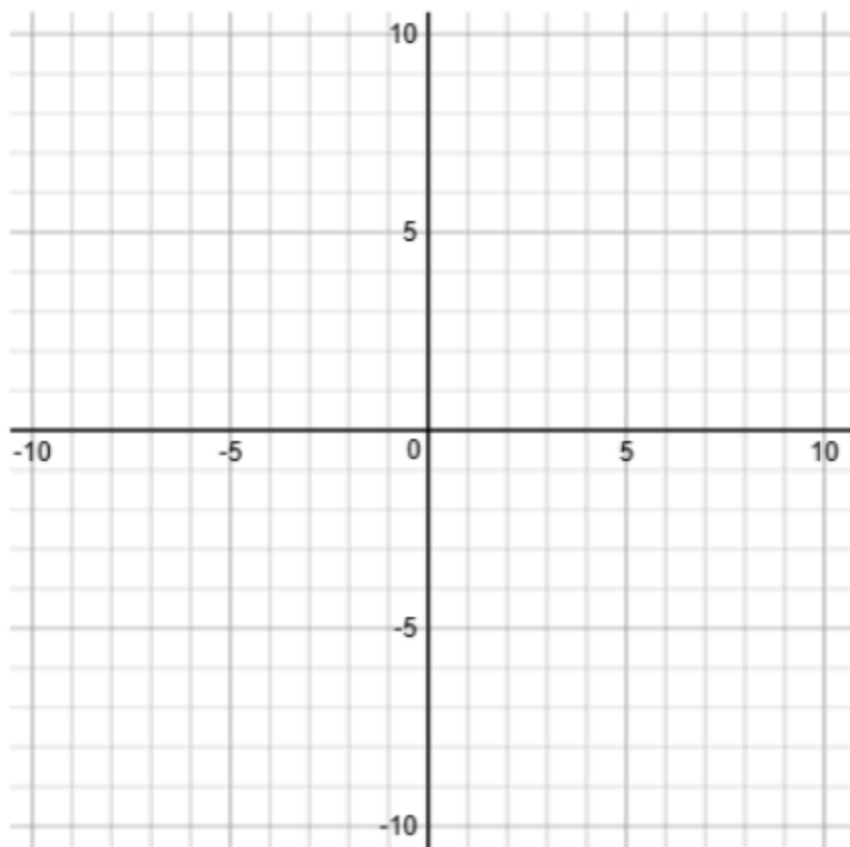
limit of $f(x)$ as x approaches 3 from the right

$\lim_{x \rightarrow 3} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 3

$$4) f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$a) f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$$b) \lim_{x \rightarrow 0^-} f(x) = \underline{\quad ? \quad}$$

$$\lim_{x \rightarrow 0^+} f(x) = \underline{\quad ? \quad}$$

$$\lim_{x \rightarrow 0} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 0 from the left

limit of $f(x)$ as x approaches 0 from the right

limit of $f(x)$ as x approaches 0